

Clinical and Behavioral Factors Associated with Alcohol Use Disorder in the All of Us Research Program

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1. Introduction, Workflow & Study Design

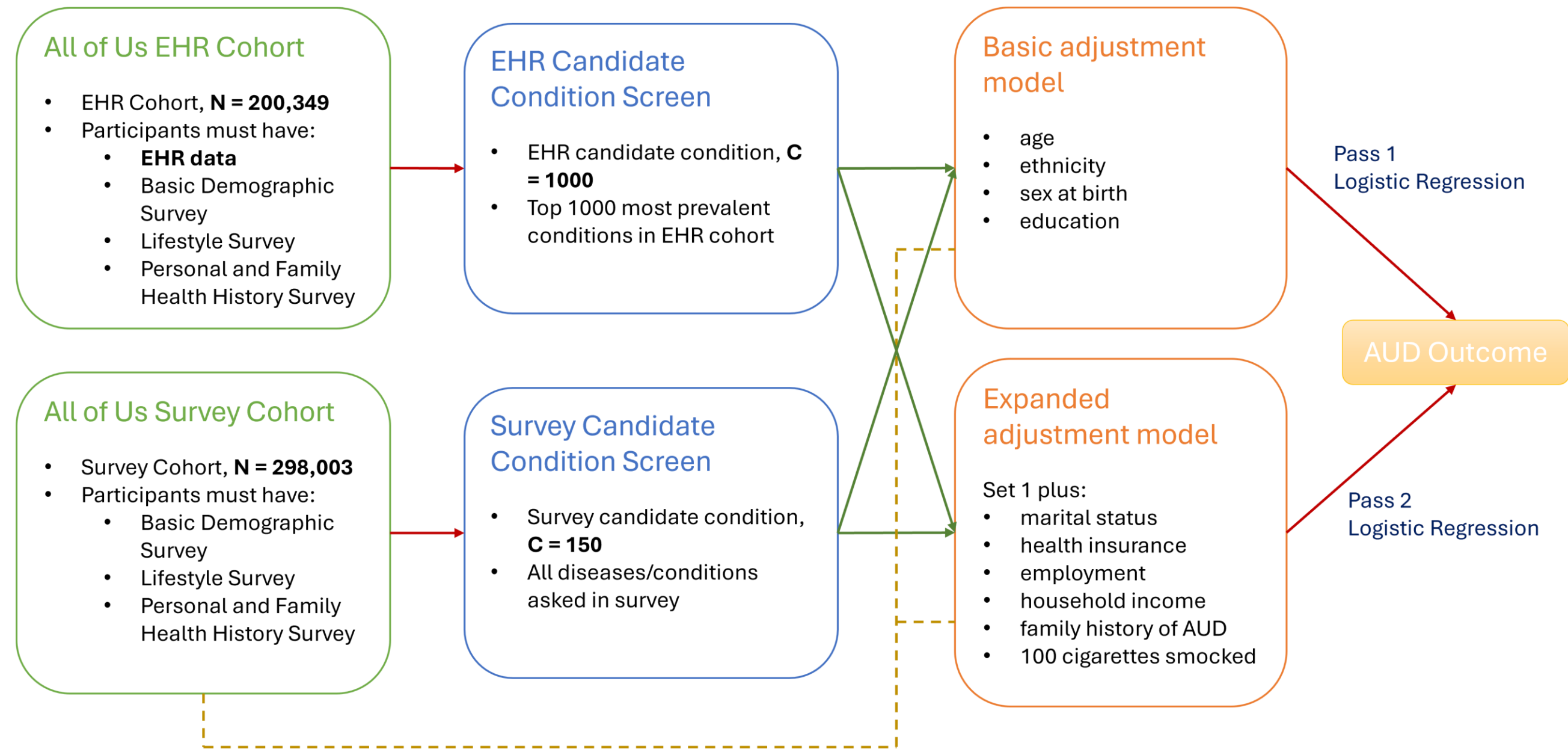
Study question

Which clinical conditions are associated with AUD after adjusting for demographic, behavioral, socioeconomic, and family-history factors?

Introduction

- Alcohol use disorder (AUD) is clinically complex and may be associated with mental health, substance use, liver disease, family history, socioeconomic background, and other life-context factors.
- The All of Us Research Program links participant survey responses with electronic health record (EHR) data, enabling large-scale analysis of AUD-associated clinical patterns.
- Survey data provide participant-reported disease history and background covariates, while EHR data provide clinically documented diagnoses.
- This study used both survey-derived and EHR-derived AUD definitions to identify clinical conditions associated with AUD after covariate adjustment.
- The main goal was to compare whether similar AUD-associated clinical themes appear across survey and EHR data.

Workflow



Method

- Data source:** All of Us Research Program survey and electronic health record (EHR) data
- AUD outcome definitions:**
 - Survey AUD: defined from the Personal and Family Health History survey
 - EHR AUD: defined from clinical diagnosis records
- Candidate condition screens:**
 - Survey screen: ~150 survey-reported diseases/conditions
 - EHR screen: top 1,000 most prevalent non-AUD EHR conditions
- Modeling approach:**
 - Used logistic regression to estimate each condition's association with binary AUD status
 - Reported adjusted odds ratios and confidence intervals
- Covariate adjustment:**
 - Basic model: age, race/ethnicity, sex at birth, education
 - Expanded model: marital status, health insurance, employment, household income, smoking history, and AUD family history
 - Education was removed from the final expanded model after losing independent significance under richer adjustment
- Survey-EHR comparison:**
 - Matched condition names across survey and EHR sources using text similarity
 - Compared adjusted odds ratios to identify convergent AUD-associated clinical signals

2. Cohort & Covariate Groups

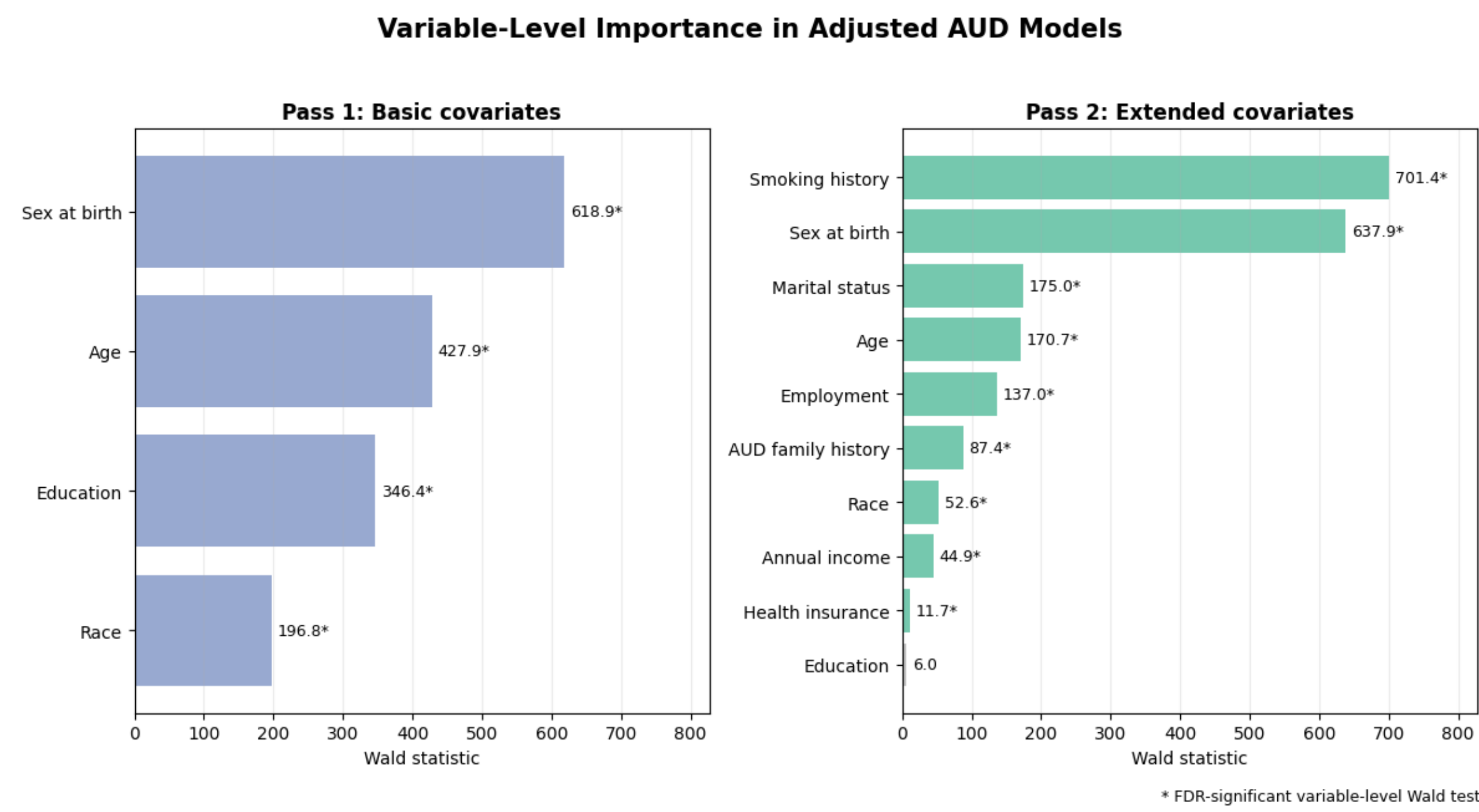
Cohort Statistics

1. Analytic Cohort Summary			
Analysis set	Sample size (N)	AUD cases (n)	AUD prevalence (%)
Basic-adjustment cohort	288,251	3,357	1.16%
Expanded-adjustment cohort	130,477	2,866	2.20%

2. Selected Participant Characteristics			
Age at baseline (years)		Sex at birth	
18-29	12.0%	Female	65.1%
30-44	21.6%	Male	34.3%
45-59	25.1%	Missing	0.7%
60-74	32.3%		
75+	8.9%		
Race		Smoking history	
White	68.9%	Never smoked 100 cigarettes	60.6%
Black or African American	9.4%	Smoked 100 cigarettes	34.4%
Other	11.0%	Missing	1.8%
More than one population	4.5%		
Asian	3.3%		
Missing	1.4%		
Education		AUD family history	
Advanced degree	29.9%	No	55.3%
College graduate	28.0%	Yes	44.7%
College 1-3 years	25.5%		
Twelve/GED	11.6%		
Less than 12th grade	3.6%		
Missing	1.4%		

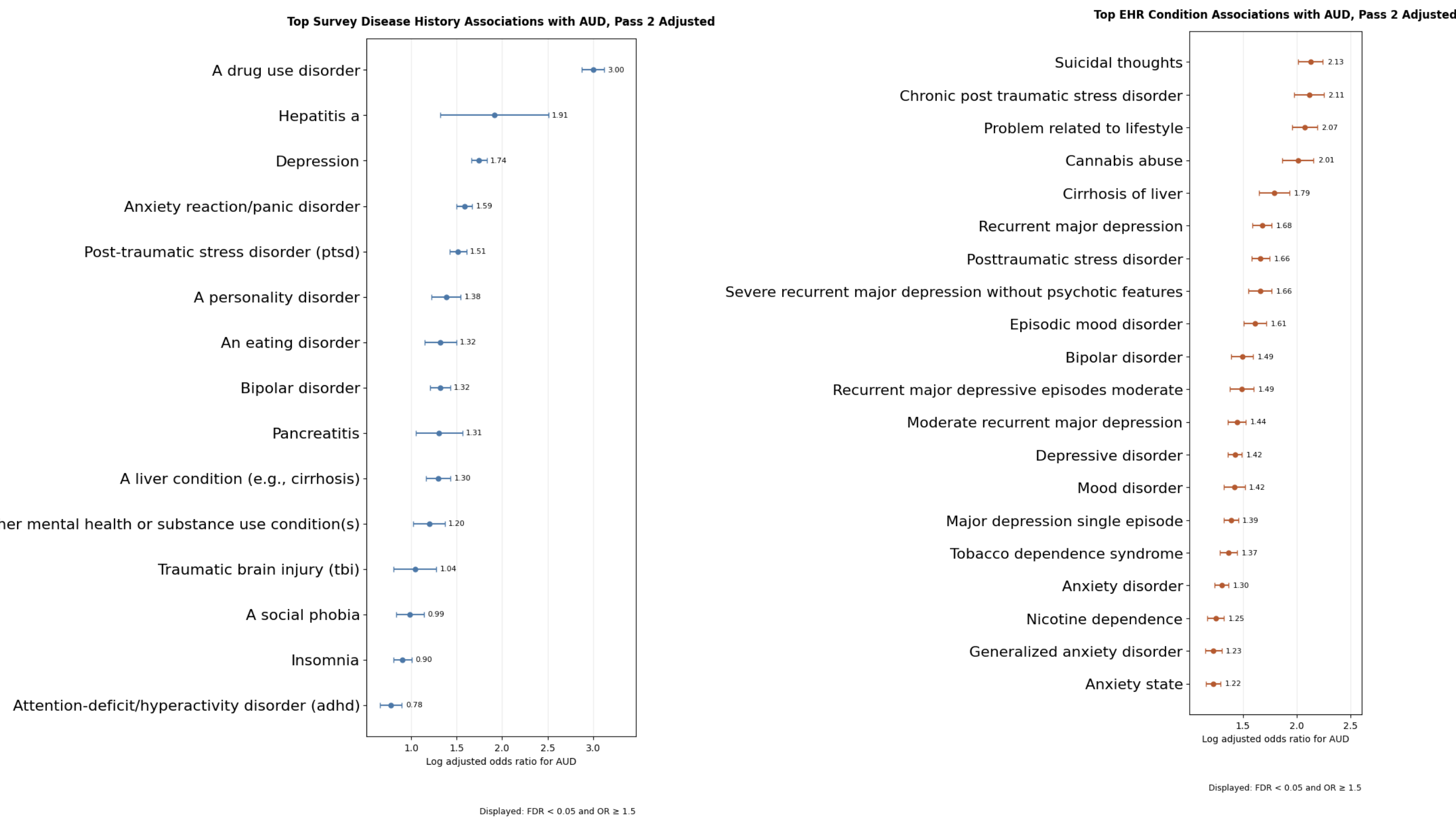
Percentages may not sum to 100% due to rounding. AUD family history percentages are among respondents.

Covariate Importance and Adjustment



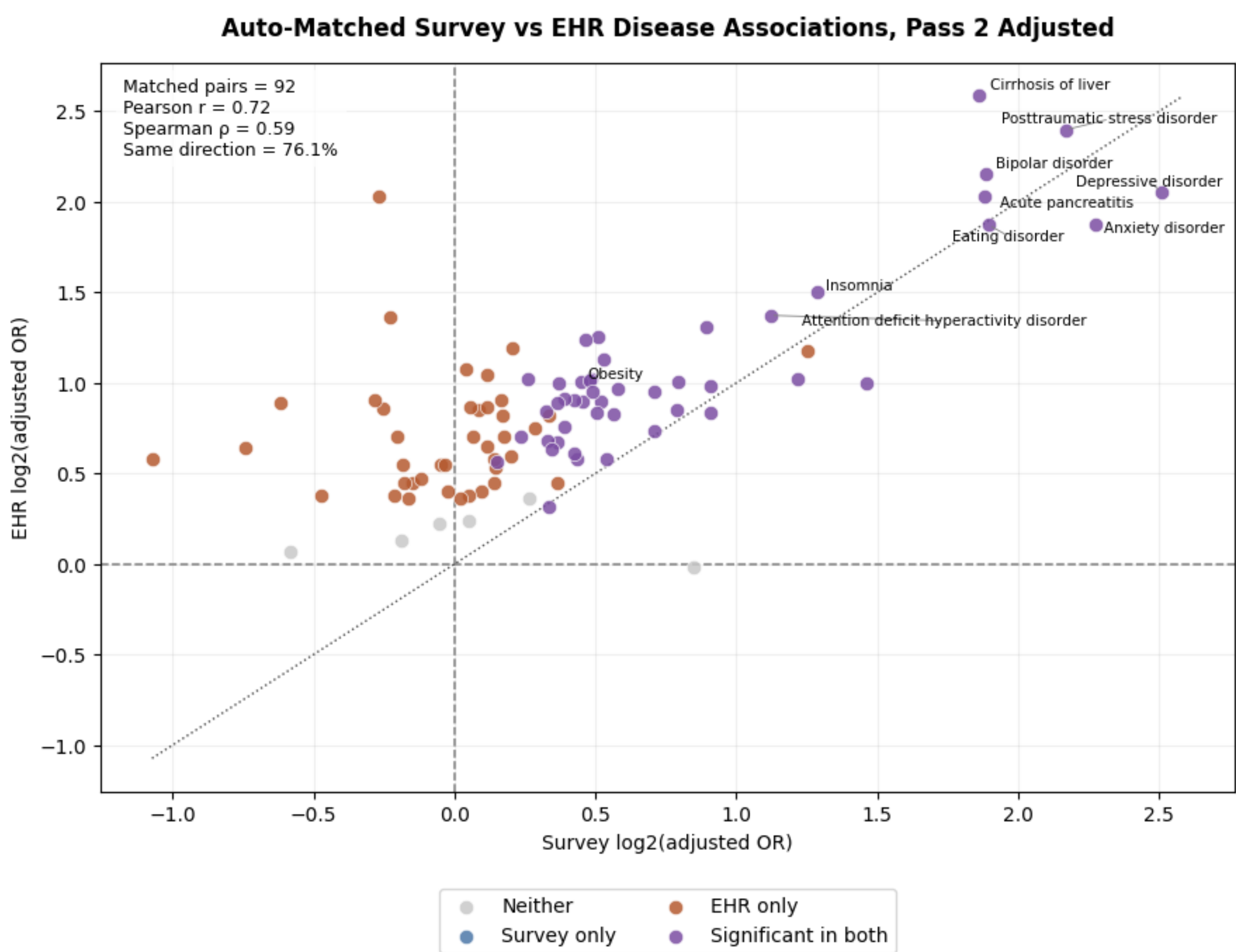
3. Main Results & Takeaway

Expanded-Adjustment Condition Associations with AUD

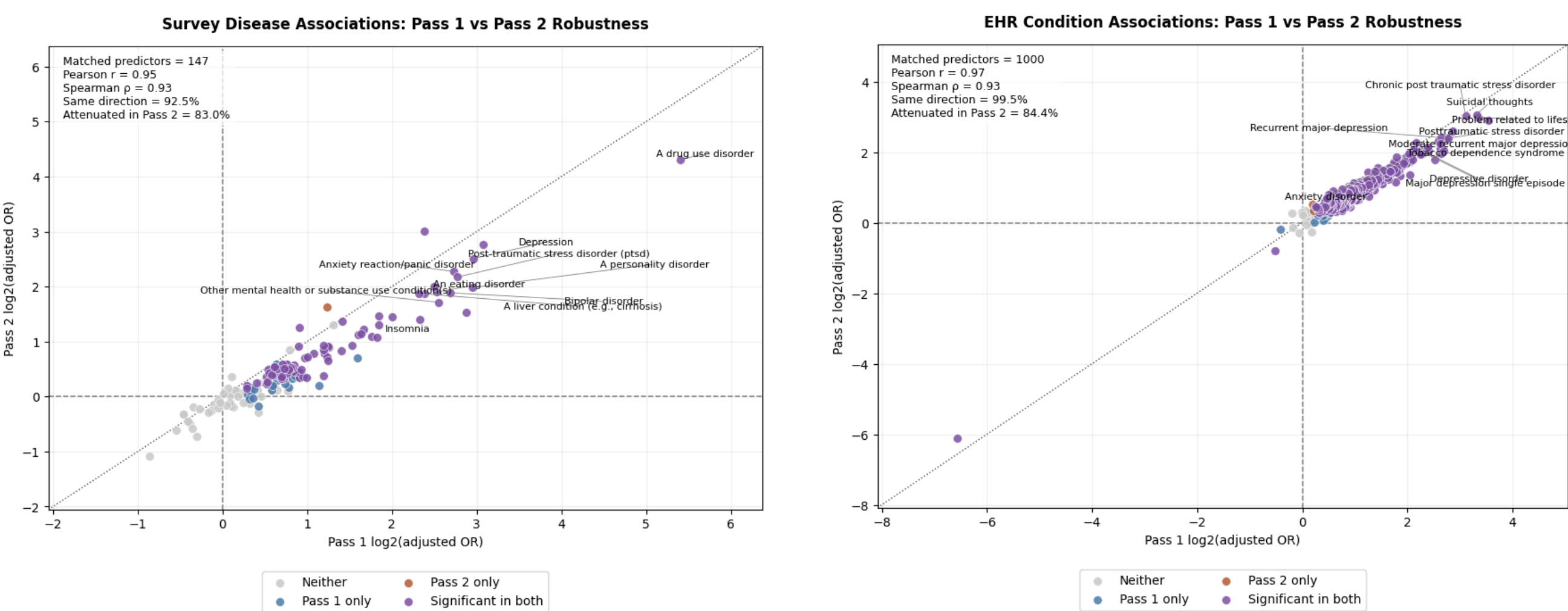


3. Main Results & Takeaway (Cont'd)

Survey-EHR Comparison: Convergent AUD Signals



Basic-Expanded Adjustment Model comparison



Conclusion

- AUD-associated conditions clustered around three major clinical domains: substance use, mental health, and liver/pancreatic disease.
- Similar clinical themes appeared in both survey-reported and EHR-derived analyses, supporting a consistent AUD-related comorbidity pattern across data sources.
- Survey and EHR data were not interchangeable: survey data captured participant-reported disease history, while EHR data reflected clinically documented diagnoses.
- Expanded covariate adjustment helped account for demographic, behavioral, socioeconomic, and family-history factors, making the observed associations more conservative and interpretable.

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